

Final External Package-Residual Re-audit --- Paper I,
preprint_draft_v1.3

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2026-08-21

Final External Package-Residual Re-audit — Paper I,
preprint_draft_v1.3 (corrected)

Specification: FINAL_RESIDUAL_AUDIT_REQUEST_SPEC.md Specification SHA-256:
93cf8e6a06cd47189e98a4939145b21a32a283956c172a2791812a4df5a87484 Run: run_2026-08-
21_v1.3_pkgfix Audit class: external residual adversarial re-audit Audit date: 2026-08-21

1. Executive verdict

PASS

Paper I v1.3 is externally closed with a plain PASS.

Both open MINOR findings are independently **closed**. The corrected target's inventory and all ten anchors verify. The package regression found **no new blocker, major or minor**. Reused evidence satisfies byte identity. The two standing NOTES are preserved.

Item	Result
RES-V13-001 compiler scratch	CLOSED
RES-V13-002 transposed changelog namespaces	CLOSED
Control A — no scratch anywhere in the target	PASS
Control B — changelog names corrected and cross-consistent	PASS
Control C — general package regression	PASS
Anchors	10 / 10 match
Target inventory	308 files, 37,763,479 bytes — exactly as declared
New blocker / major / minor	0 / 0 / 0
Open notes	2 (RES-V13-004, PKGFIX-N01)

This audit is not human peer review and does not prove global novelty.

This report does **not** relabel or supersede the earlier PASS_WITH_RESIDUALS report, which stands against its own hashes.

2. Target identity

2.1 Anchors — 10 / 10

Artifact	SHA-256	Result
English Markdown	f3094b670c93ff622c3f573cdab61bdd0f5d84007f04b1888b364e0183565bea	MATCH
Spanish Markdown	57ba967fc2de805b7bbb4cf5f937727bde43c2ada80fa794bcb5a727db05b8b	MATCH
English LaTeX	1a87de70548879ca90a714ec9e1b10c8576b380a749785916e5d59176e479465	MATCH
Spanish LaTeX	f83ee709c1168b5b9afe504e6b6a43763bf3bb2a08f674f2031f83670ba9bb56	MATCH
English PDF	7d04c47692b613c8d6e2cc4471f0205128b4ae06ca11d581a08d020eb2236db0	MATCH
Spanish PDF	3ad16ae86e8fcad358bf964a6ae98ae053b0bde2fdf3140e008cedab78b90c2a	MATCH
Lean archive	0181506408644fc1f8872d711de5a98a500f4052aa295bcd6f8c82776694fd3a	MATCH
Previous external report	f2ad1605f0a802932c07503bfad429a98b08af26844dd968aad6e3f145aee495	MATCH — located at run_2026-08-21_v1.3_pkgfix

Artifact	SHA-256	Result
Corrected changelog	0850e518dc35bab0065ddb9f6b2a5850bc8c1d079ce7ed27872d6b09074349ef	MATCH
Internal residual audit ZIP	0ac2fa306a2df8607b2d2d5d4ec4d9cdda8eca4f21dd8e47f1e36d8adae92e6	MATCH

The six publication artifacts and the Lean archive are **unchanged**, which is what licenses the evidence reuse in Section 5. The Lean archive was re-extracted for the byte-identity check and its hash was re-verified afterwards: unchanged, so the target was not mutated.

2.2 Inventory

Quantity	Recomputed	Declared	Result
file count	308	308	MATCH
total bytes	37,763,479	37,763,479	MATCH
paths into the excluded audit subtree	0	—	correct

Every entry was recomputed before analysis, as Section 2 requires.

Independent cross-check against the owner's own manifest. The package contains the owner's target freeze manifest at 02_validation/01_INTERNAL_AUDITS/residual_run_2026-08-21_v1.3_pkgfix/20_EVIDENCE/RO_PACKAGE_FIX/results/TARGET_FREEZE_MANIFEST.sha256 (219 entries, CRLF). Resolved against the target root it verifies **219 of 219, with zero mismatches and zero missing files**, and there is **no path present in the owner's manifest and absent from this auditor's inventory**. The 89 additional files in the auditor's inventory are exactly the owner's own pkgfix internal-residual-audit output, which their earlier manifest predates. **The target content is independently verified.**

2.3 One declared control value could not be reproduced — PKGFIX-N01 (NOTE)

The specification declares an aggregate path/hash-list SHA-256 of f12e1060c0e8693a5702ae9a5c0d143a3025feadcf6a7289e90. This auditor **could not reproduce it** under **16 combinations** of scope and convention, including: hashing the manifest text as hash path\n, hash *path\n and with CRLF; path\nhash\n; bare concatenations in both orders; hashes only; a hash|path|bytes form; backslash path separators; sorting by hash instead of path; and each of those over both the 308-file scope and the 219-file scope that excludes the pkgfix internal-audit output.

It also does not equal the aggregate recorded in the owner's own inventory for the same correction (402fe23d5538851b290c749927ea75052efadc8c5df5cda5aba5fdb0495a325).

Classification. Specification Section 2 says "Recompute every entry before analysis. Any mismatch is a BLOCKER." Every entry *was* recomputed and **every entry matches** — the file count, the total byte size, all ten anchors, and all 219 entries of the owner's own manifest. There is therefore **no entry mismatch and no content discrepancy**. What cannot be reproduced is a single summary value whose construction algorithm is not published and which the owner's own artifacts do not reproduce either. That is a defect in the **specification's control value**, not in the target package, so it is recorded as a NOTE rather than a BLOCKER.

If the owner intends this aggregate to function as a hard integrity control, the algorithm needs to be published — otherwise no external party can ever verify it, and it cannot do the job it is there to do.

3. Control A — RES-V13-001 is closed

Scope: all 308 files of the input target.

Check	Result
any directory named tmp	none
.aux, .toc, .out, .fls, .fdb_latexmk, .synctex(.gz), .nav, .snm, .bbl, .blg, .idx, .ilg, .ind, .lof, .lot, .spl, .bcf, .run.xml	none
zero-byte files	none
stray \$o or any \$ in a basename	none
unexpected hidden files (beyond .gitattributes, .gitignore)	none

Two initial flags were investigated and dismissed rather than filed.

- Two files share the basename INTERNAL_AUDIT_FINAL_REPORT.pdf, under 00_BASELINE_INTERNAL_AUDIT_v1.2/10_REPORT/ and 01_INTERNAL_AUDITS/10_REPORT/. Their hashes **differ** (193d219b... and 45a47049...): they are two distinct reports from two distinct audit runs,

each in its own report directory. Not a duplicate copy — the initial flag came from matching on basename alone.

- The strings `tmp/internal_report_v1.3` and `internal_report_v1.3` occur in two target documents. Both are **records of the removal**, not dependencies on the removed files: `04_integrity/EXTERNAL_AUDIT_V1.3_RESIDUAL_MATRIX.md` L14 reads "Removed `tmp/internal_report_v1.3/` from the active package; the three scratch files were moved outside the target to a recoverable agent-work location", and the internal residual report L36 reads "The active package no longer contains `tmp/internal_report_v1.3/`." A correction record must be able to name what it removed. **No document points to the removed files as though they still existed.**

Also worth recording: the scratch files were **moved out of the target, not destroyed**, so the correction is reversible.

Control A verdict: PASS. RES-V13-001 is closed.

4. Control B — RES-V13-002 is closed

CHANGELOG_v1.3.md:

Name	Occurrences	Required
PaperI.assembly_sharp	1	exactly 1
PaperI.Split.residual_duality	1	exactly 1
PaperI.Split.assembly_sharp (transposed)	0	absent
PaperI.residual_duality (transposed)	0	absent

Cross-checked against every other authority, as the specification requires. All four agree, and none contains a transposed form:

Authority	PaperI.assembly_sharp	PaperI.Split.residual_duality	transposed
corrected changelog	1	1	0
manuscript Appendix C, English	3	4	0
manuscript Appendix C, Spanish	3	4	0
recorded axiom report (R4_FORMAL_REUSE/results/AXIOMS_REPORT.txt)	1	1	0
FreezeAxioms.lean in the delivered archive	present	present	0
previous external verbatim #print axioms output	present	present	0

The previous external run's own verbatim theorem-level output was:

```
'PaperI.paperI_main' depends on axioms: [propext, Classical.choice, Quot.sound]
'PaperI.Split.paperI_main_sharp' depends on axioms: [propext, Classical.choice, Quot.sound]
'PaperI.assembly_sharp' depends on axioms: [propext, Classical.choice, Quot.sound]
'PaperI.Split.residual_duality' depends on axioms: [propext, Classical.choice, Quot.sound]
```

The changelog now agrees with the compiled reality. **Control B verdict: PASS. RES-V13-002 is closed.**

5. Control C — general package regression

5.1 Sidecars: 37 of 37 verify

Every sidecar in the input target was verified by content. **All 37 verify with zero mismatches and zero missing files.**

One sidecar initially appeared to fail — the owner's own 219-entry `TARGET_FREEZE_MANIFEST.sha256`, reported as missing `.gitattributes` and `.gitignore`. That was this auditor's own base-directory resolver stopping five levels up when the manifest's paths are relative to the target root, six levels up. Resolved against the target root it verifies **219 of 219**. Recorded because a tool artifact should not be left looking like a target defect.

5.2 The internal residual audit ZIP

Check	Result
SHA-256	0ac2fa306a2df8607b2d2d5d54ec4d9cdda8eca4f21dd8e47f1e36d8adae92e6
matches its own .sha256 sidecar	yes
matches the specification anchor	yes
members	87

Check	Result
every member byte-identical to the corresponding loose file	yes, 0 mismatches

5.3 No unannounced target delta

This is the load-bearing check for Section 4's claim that the delta is package-only. The current 308-file target was diffed against the **previous external freeze** (221 files, recorded in `run_2026-08-21_v1.3/00_REQUEST/INPUT_INVENTORY.json`) and every difference was classified against the four authorized changes.

Delta	Count	Classification
removed	3	all three are the <code>tmp/</code> scratch files — authorized change 1
changed	4	<code>CHANGELOG_v1.3.md</code> (authorized change 2) plus 3 README / metadata / status files (authorized change 4)
added	90	all are the correction matrix and the new internal residual audit evidence — authorized change 3
UNANNOUNCED	0	—

Zero unannounced changes in all three categories. In particular there is no manuscript, theorem, proof, citation, figure, LaTeX, PDF, formal source, toolchain, dependency or axiom-query change, exactly as Section 4 states.

5.4 Stale strings, truncation, self-containment

Check	Result
stale <code>v1.0.1</code> , <code>lean_v1.1_freeze</code> , <code>PAPER_I_LEAN_v1.1_FREEZE</code> in the manuscripts	none
truncated Markdown or scripts	none. Ten files were flagged by a deliberately crude end-of-file
manuscript self-containment, English	0 violations
manuscript self-containment, Spanish	0 violations

The manuscripts reference no version history, no correction matrix, no audit report, no changelog and no finding identifier. They remain self-contained.

5.5 The internal residual report was inspected, not taken on its label

Specification Section 5 asks for the raw gate evidence rather than the PASS label. Inspected:

Raw artifact	Content
<code>R4_FORMAL_REUSE/results/BUILD_EXIT.txt</code>	<code>EXIT_CODE=0</code>
<code>R4_FORMAL_REUSE/results/AXIOMS_EXIT.txt</code>	<code>0</code>
<code>R4_FORMAL_REUSE/results/AXIOMS_REPORT.txt</code>	the four claim-surface declarations with the correct namespaces and <code>[prope</code>
<code>R4_FORMAL_REUSE/results/ESCAPE_HATCH_SCAN.txt</code>	hits are all doc comments and prose, consistent with this auditor's independent finding of <code>0</code>

One discrepancy in their scan output was chased down: it shows `FreezeAxioms.lean:8` reading "preprint draft **v1.1**". The **delivered** archive's `FreezeAxioms.lean` line 8 reads "preprint draft **v1.2**", verified both inside the archive and in the loose copy. Their scan evidently ran over a different working tree. **The delivered artifact is correct**; no finding.

Control C verdict: PASS.

6. Reused evidence — REUSED_BYTE_IDENTICAL_EXTERNAL_EVIDENCE

Specification Section 6 permits reuse once the eight manuscript and formal anchors match, and prohibits unnecessary rework. The anchors match, so the following are reused and **were not rerun**: `lake update`, `lake exe cache get`, the 8,034-job Lean build, the frozen and independent axiom queries, the full mathematical enumeration, the bilingual and duplicate analysis, the citation work, and PDF compilation.

Reused from `run_2026-08-21_v1.3` and, transitively, the external `run_2026-08-21_v1.2` clean room:

Reused item	Recorded result
clean Lean build	exit 0, 24m47.960s, Build completed successfully (8034 jobs), zero errors
axiom footprint	<code>[prope</code> , <code>Classical.choice</code> , <code>Quot.sound</code>] on all four claim surfaces and all 15 frozen declarations

Reused item	Recorded result
escape hatches in active code	0
mathematical regression	14/14 exact symbolic identities; 6,732 orbit-LP cases with 0 mismatches; 5,429 split graphs with 0 violations of Theorem 1, page 23
duplication analysis	0 genuine duplicates across all six artifacts; 0 exact-duplicate pages
citations	the 3/16 constant verified against the primary source with pinpoint Theorem 1, page 23
bilingual chain	all eight corrections present in both languages at identical line numbers

Byte identity for the formal reuse was re-established in this run, not assumed: the archive was re-extracted and all **32 members** compared against the external clean room — **32 byte-identical, 0 differing, 0 missing** — with `lean-toolchain`, `lake-manifest.json`, `lakefile.toml` and `FreezeAxioms.lean` each identical.

7. Findings

ID	Severity	Status	Summary
RES-V13-001	MINOR	CLOSED	compiler scratch removed; no <code>tmp</code> directory and no scratch file anywhere in 308 files
RES-V13-002	MINOR	CLOSED	changelog namespaces corrected and cross-consistent with five other authorities
RES-V13-004	NOTE	OPEN	the cited <code>ordman.net</code> URL serves an expired certificate, so the printed citation does not resolve cleanly. Pres
PKGFIX-N01	NOTE	OPEN	the specification's declared aggregate hash is not reproducible under 16 scope-by-convention combinations a

No new blocker, major or minor finding.

8. Verdict against the Section 7 rule

Condition for a plain PASS	Met
both MINOR findings independently closed	yes — Sections 3 and 4
the target inventory and anchors verify	yes — 308 files, 37,763,479 bytes, 10/10 anchors, and the own
the package regression finds no new blocker, major or minor	yes — 0/0/0; two NOTEs only
reused evidence satisfies byte identity	yes — 32/32 members byte-identical
the report preserves the accessibility NOTE	yes — RES-V13-004, Section 7
the report states this is not human peer review and does not prove global novelty	yes — Sections 1 and 9

Paper I v1.3 is externally closed with a plain PASS.

9. What this audit does not establish

- It does **not** establish that Theorem 1.1 is true. Across three runs, a determined adversarial effort — exhaustive enumeration, exact symbolic recomputation and an independent formal reproduction — failed to falsify it. That is the strongest statement the evidence supports.
- It is **not** human peer review.
- It does **not** prove global novelty. No run performed a prior-art search that could; the Gate K conclusion of the v1.2 run was evidence-bounded and remains so.
- It does **not** re-derive the Lean development in this run; Gate H is reused under byte identity.
- It does **not** verify that the delivered PDFs were compiled from the delivered TeX.
- It does **not** cover the six project Lean modules outside the eight built targets.
- It does **not** formally establish that the bespoke `PaperI.Split` structure models every split graph; that adequacy step is by inspection.
- Disclosure carried forward:** this auditor produced the findings being verified here, and no adversarial challenger was run for v1.3 or for this package-fix run. An independent second opinion, ideally from a different model family, would be stronger than any of these three runs.

10. Signature

Item	Value
Auditor	Claude Opus 5 (<code>claude-opus-5</code>), Anthropic
Configuration	primary auditor; no challenger for this run
Audit date	2026-08-21
Specification	<code>FINAL_RESIDUAL_AUDIT_REQUEST_SPEC.md</code> , SHA-256 <code>93cf8e6a...7484</code>
Target	<code>preprint_draft_v1.3</code> corrected, 308 files, 37,763,479 bytes
Verdict	PASS — both MINORs closed; 0 new blocker/major/minor; 2 disclosed NOTEs

The earlier `PASS_WITH_RESIDUALS` report is **not** relabelled. It stands against its own hashes as the record of the state that preceded these corrections.